A CRITERION FOR A SITE TO HAVE ENOUGH POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The criterion for a localized site to have enough points. Same chapter, Proposition proposition-criterion-points. Supplies the hypothesis on the localized sites and the conclusion there.
- Points of a localized site give points of the site. Same chapter, Lemma lemma-point-morphism-sites. Turns the points of the localized sites into points of the ambient site.

Source reference. Original proposition 00YQ: sources/stacks-sites/pinned-original-source.tex, source lines 9663–9663. Navigation pointer invoked at proof line 9663; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05V1: sources/stacks-sites/pinned-original-source.tex, source lines 9671–9671. Navigation pointer invoked at proof line 9671; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let I be a set and for $i \in I$ let U_i be an object of $\mathcal{C}$ such that*

(1) *$\coprod h_{U_i}$ surjects onto the final object of $Sh(\mathcal{C})$, and*
(2) *$\mathcal{C}/U_i$ satisfies the hypotheses of Proposition 00YQ.*

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Then $\mathcal{C}$ has enough points.

Proof. By assumption (2) and the proposition $\mathcal{C}/U_i$ has enough points. The points of $\mathcal{C}/U_i$ give points of $\mathcal{C}$ via the procedure of Lemma 05V1. Thus it suffices to show: if $\phi : \mathcal{F} \to \mathcal{G}$ is a map of sheaves on $\mathcal{C}$ such that $\phi|_{\mathcal{C}/U_i}$ is an isomorphism for all i, then ϕ is an isomorphism. By assumption (1) for every object W of $\mathcal{C}$ there is a covering $\{W_j \to W\}_{j \in J}$ such that for $j \in J$ there is an $i \in I$ and a morphism $f_j : W_j \to U_i$. Then the maps $\mathcal{F}(W_j) \to \mathcal{G}(W_j)$ are bijective and similarly for $\mathcal{F}(W_j \times_W W_{j'}) \to \mathcal{G}(W_j \times_W W_{j'})$. The sheaf condition tells us that $\mathcal{F}(W) \to \mathcal{G}(W)$ is bijective as desired. □